	Raffles Institution Physics Department Revision Package 3	Nuclear Physics	20

***Important note:** *denotes questions that may require knowledge from later chapters in order to solve them.*

I. MCQs (ACJC – YJC)

1. [ACJC/H2/Prelims 2018/P1/29]

Uranium-238 undergoes a series of radioactive decay to form various daughter products.

X is produced after uranium-238 emitted 4 alpha and 2 beta decays.

What is the number of neutrons and protons in X?

	Neutron number	Proton number
A	136	86
B	136	94
C	222	86
D	222	94

2. [ACJC/H2/Prelims 2018/P1/30]

After 100 days, 10 % of a radioactive sample of $^{232}_{90}Th$ has decayed.

How many half-lives have elapsed?

A 0.15 B 0.20 C 3.3 D 660

3. [AJC/H2/Prelims 2018/P1/29]

Which equation shows a radioactive decay that emits an alpha particle?

A $^{14}_{7}N + ^{1}_{1}p \rightarrow ^{11}_{6}C + X$

B $^{220}_{86}Rn \rightarrow ^{216}_{84}Po + X$

C $^{137}_{55}Cs \rightarrow ^{137}_{56}Ba + X$

D $^{60}_{28}Ni \rightarrow ^{60}_{28}Ni + X$

4. [AJC/H2/Prelims 2018/P1/30]

Which of the following statements concerning nuclear properties is true?

A The greater the binding energy of a nucleus, the more stable it is.

B If the total rest mass of the products of a reaction is greater than the total rest mass of the reactants, this reaction is impossible.

C When a stationary nucleus decays by emitting a γ photon, the daughter nucleus will move off in opposite direction to the γ photon.

D The half-life of a radioactive isotope can be changed by allowing the isotope to react chemically to produce a new compound.

5. [CJC/H2/Prelims 2018/P1/29]

A stationary uranium nucleus of mass 238 units disintegrates by the emission of an α-particle of mass 4 units.

The ratio $\dfrac{\text{kinetic energy of the } \alpha\text{-particle}}{\text{kinetic energy of the recoiling daughter nucleus}}$ is

A $\dfrac{4}{234}$ B $\dfrac{4}{238}$ C $\dfrac{234}{4}$ D $\dfrac{238}{4}$

6. [CJC/H2/Prelims 2018/P1/30]

The half-life of a certain radioactive material is 3.0 s.

How long does it take for its activity to become 10 % of the original activity?

A 0.46 s B 5.4 s C 10 s D 15 s

7. [DHS/H2/Prelims 2018/P1/29]

Two deuterium nuclei fuse together to form a Helium-3 nucleus, with the release of a neutron. The reaction is represented by

$$^{2}_{1}H + ^{2}_{1}H \rightarrow ^{3}_{2}He + ^{1}_{0}n + \text{energy}$$

The binding energies per nucleon are:

$^{2}_{1}H$ 1.09 MeV,

$^{3}_{2}He$ 2.54 MeV.

How much energy is released in this reaction?

A 0.36 MeV B 1.45 MeV C 3.26 MeV D 5.44 MeV

8. [DHS/H2/Prelims 2018/P1/30]

Nuclide X decays to stable nuclide Y with a half-life of T years.

Geologists are sure that nuclide Y found in a particular rock sample has all came from nuclide X which was present when the rock formed.

The rock is thought to be $3T$ years old.

What is the expected ratio $\dfrac{\text{number of atoms of X}}{\text{number of atoms of Y}}$ for this rock?

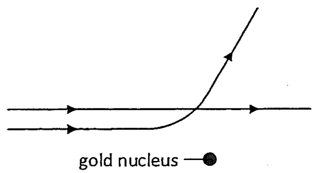
A 1/6 B 1/7 C 1/8 D 1/9

9. **[EJC/H2/Prelims 2018/P1/28]**

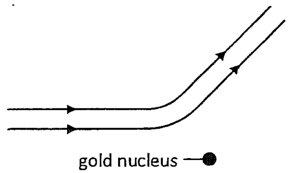
Two alpha particles with equal energies are fired towards the nucleus of a gold atom.

Which diagram best represents their paths?

A



B



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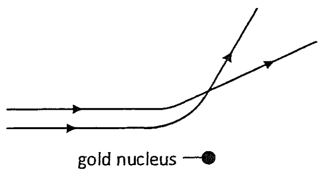
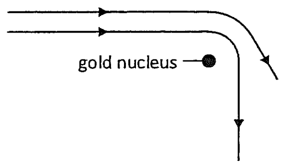


D



10. **[EJC/H2/Prelims 2018/P1/29]**

Uranium-235 undergoes fission in a reaction shown below, releasing 195 MeV of energy. The binding energy per nucleon for uranium-235 is 7.6 MeV, and those for caesium and rubidium are approximately **X** MeV.

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{143}_{55}Cs + ^{90}_{37}Rb + 3^{1}_{0}n$$

What is the value of **X**?

A 6.7 B 6.8 C 8.4 D 8.5

11. **[ACJC/H2/Prelims 2018/P1/30] [JJC/H2/Prelims 2018/P1/30]**

Radon $^{222}_{86}Ra$ is the start of a decay chain that forms bismuth $^{214}_{83}Bi$ by α and β emission.

For the decay of each nucleus of radon, how many α-particles and β-particles are emitted?

	α-particles	β-particles
A	1	1
B	2	1
C	1	2
D	2	2

12. **[HCI/H2/Prelims 2018/P1/29]**

Which of the following combinations of radioactive decay results in the formation of an isotope of the original nucleus?

A one α and four β decays

B one α and two β decays

C two α and two β decays

D four α and one β decays

13. **[HCI/H2/Prelims 2018/P1/30]**

A radioactive source consists of a mixture of two isotopes P and Q.

P has a half-life of 60 minutes and Q has a half-life of 30 minutes. The initial activity recorded by a suitable counter is 800 min^{-1}. After 120 minutes, the counter registers an activity of 80 min^{-1}.

What is the initial contribution of P to the count rate?

A 160 min^{-1} B 240 min^{-1} C 270 min^{-1} D 480 min^{-1}

14. **[IJC/H2/Prelims 2018/P1/29]**

$^{238}_{92}U$ decays through a series of transformations to a final stable nuclide. The particles emitted in the successive transformations are

$$\alpha \rightarrow \beta \rightarrow \beta \rightarrow \alpha \rightarrow \alpha$$

Which nuclide is not produced during this series of transformations?

A $^{228}_{88}Ra$ B $^{230}_{90}Th$ C $^{234}_{91}Pa$ D $^{234}_{92}U$

15. **[IJC/H2/Prelims 2018/P1/30]**

The equation

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{121}_{45}Rh + ^{113}_{47}Ag + 2^{1}_{0}n$$

shows the fission of a uranium–235 nuclide by a slow-moving neutron into a rhodium-121 nuclide, a silver-113 nuclide and two neutrons.

binding energy per nucleon of $^{235}_{92}U$ = 7.59 MeV

binding energy per nucleon of $^{121}_{45}Rh$ = 8.26 MeV

binding energy per nucleon of $^{113}_{47}Ag$ = 8.52 MeV

What is the energy released during this fission process?

A 9.19 MeV B 24.4 MeV C 73.9 MeV D 179 MeV

16. [JJC/H2/Prelims 2018/P1/29]

The activity of a sample of carbon taken from an archaeological specimen was 190 counts per minute. The activity of an equal mass of carbon taken from a living plant was 360 counts per minute. The background count was 20 counts per minute.

Taking the half-life of carbon-14 to be 5700 years, what was the approximate age of the archaeological specimen?

A 2850 year

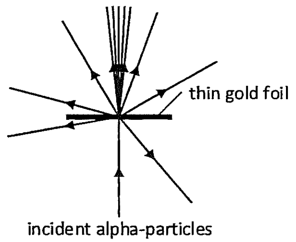
B 5700 years

C 10800 years

D 11400 years

17. [MI/H2/Prelims 2018/P1/29]

The diagram shows a thin gold foil bombarded with alpha-particles.



What does the results of this experiment provide?

A wave properties of a gold atom

B size of a gold nucleus

C mass number of a gold atom

D binding energy of a gold nucleus

18. [MI/H2/Prelims 2018/P1/30]

A radioactive sample of half-life of 10 minutes is placed 40.0 cm away from a radioactivity detector. The detector gives an average count-rate of 39.0 s^{-1}. In the absence of the radioactive sample, the detector records an average count-rate of 5.0 s^{-1}.

After 20 minutes, the detector, which is still facing the radioactive sample, is moved 20.0 cm nearer the radioactive sample. The sample can be regarded as a point source of radiation.

What is the average count rate on the detector?

A 8.5 s^{-1} **B** 13.5 s^{-1} **C** 34.0 s^{-1} **D** 39.0 s^{-1}

19. [MJC/H2/Prelims 2018/P1/28]

When the number of protons and the number of neutrons in a nuclide are both "magic numbers", it is more stable than expected. Such nuclides are termed "doubly magic".

The first few "magic numbers" are 2, 8, 20, 28, 50, 82, and 126.

$^{28}_{8}O$ $^{40}_{20}Ca$ $^{56}_{26}Fe$ $^{50}_{28}Ni$ $^{126}_{50}Sn$

How many of the above nuclides are "doubly magic"?

A 1 **B** 2 **C** 3 **D** 4

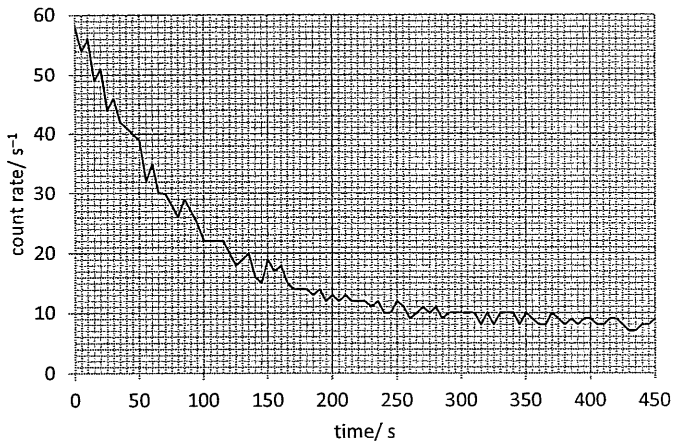
20. [MJC/H2/Prelims 2018/P1/29]

Radon-222, $^{222}_{86}Rn$ decays to lead-210, $^{210}_{82}Pb$ via a series of three alpha and two beta decays through a series of intermediate nuclides. Which of the following cannot be one of the intermediate nuclides produced?

A $^{214}_{82}Pb$ **B** $^{214}_{83}Bi$ **C** $^{218}_{84}Po$ **D** $^{216}_{85}At$

21. [MJC/H2/Prelims 2018/P1/30]

An experiment is carried out in which the count rate is measured at a fixed distance from a sample of a certain radioactive material. The figure below shows the variation of count rate with time.



What is the approximate half-life of the material?

A 60 s **B** 80 s **C** 100 s **D** 120 s

22. [NJC/H2/Prelims 2018/P1/28 Modified]

The number of radioactive nuclides in two different samples *P* and *Q* are initially 4*N* and *N* respectively. If the half-life of *P* is *t* and that of *Q* is 2*t*, the number of radioactive nuclei in *P* will be the same as the number of radioactive nuclei in *Q* after a time of

A $t/2$ **B** $2t$ **C** $4t$ **D** $8t$

23. [NJC/H2/Prelims 2018/P1/29]

A parent nucleus, initially at rest, decays into two particles of masses m_1 and m_2, moving away from each other in opposite directions. If the decay releases energy *E*, what is the kinetic energy of mass m_1?

A $\frac{m_1}{m_2}E$ **B** $\frac{m_2}{m_1}E$ **C** $\frac{m_2}{m_1+m_2}E$ **D** $\frac{m_1}{m_1+m_2}E$

24. **[NJC/H2/Prelims 2018/P1/30]**

Hydrogen bombs operate with tritium, a radioactive isotope of hydrogen with a half-life of 1.7×10^8 s. It has been proposed that if the production of tritium were halted, countries storing hydrogen bombs would have to replenish supplies from existing bombs. Eventually there would not be enough tritium to make any bombs.

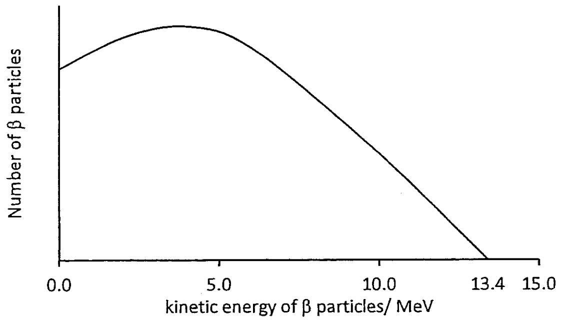
A country has a stockpile of 2000 hydrogen bombs.

What is the shortest duration that the world had to wait for the country to be unable to make a single bomb?

A 54 years **B** 59 years **C** 10800 years **D** 1.7×10^9 years

25. **[NYJC/H2/Prelims 2018/P1/28]**

The beta spectrum for ^{12}B decay is as shown below.



The kinetic energy of an emitted β-particle is 6.0 MeV.

What is the approximate energy of the associated neutrino?

A 4.0 MeV **B** 6.0 MeV **C** 7.4 MeV **D** 13.4 MeV

26. **[NYJC/H2/Prelims 2018/P1/29]**

Two radioactive isotopes P and Q have half-lives of 10 minutes and 15 minutes respectively. A sample initially contains the same number of atoms of each isotope.

After 30 minutes, the ratio of the number of atoms of P to the number of atoms of Q will be

A 0.25 **B** 0.50 **C** 1.0 **D** 2.0

27. **[NYJC/H2/Prelims 2018/P1/30]**

At time t, a sample of a radioactive substance contains N atoms of a particular nuclide.

At time $t+\Delta t$, where Δt is a short period of time, the number of atoms of the nuclide is $N-\Delta N$.

Which of the following expressions gives the decay constant of the nuclide?

A $\dfrac{\Delta N}{N}$ **B** $\dfrac{N-\Delta N}{t+\Delta t}$ **C** $\dfrac{\Delta N}{N\Delta t}$ **D** $\dfrac{N\Delta N}{\Delta t}$

28. **[PJC/H2/Prelims 2018/P1/29]**

A radioactive nuclide X disintegrates by emitting gamma radiation and a single α particle, forming a daughter nuclide Y.

Which of the following statements is correct?

A X has more protons in its nucleus than Y.

B X and Y are isotopes of the same element.

C The atomic number of X is less than that of Y.

D The mass number of X is one less than that of Y.

29. **[PJC/H2/Prelims 2018/P1/30]**

A Geiger-Muller tube recorded an average count-rate of 20 min^{-1} in the absence of any radioactive source. When it is moved near a radioactive source of half-life 48 hours, the average count-rate rises to 100 min^{-1}.

What is average count rate recorded of the source 12 hours later?

A 67 min^{-1} **B** 84 min^{-1} **C** 87 min^{-1} **D** 94 min^{-1}

30. **[RVHS/H2/Prelims 2018/P1/29]**

Radioactive decay is both spontaneous and random in nature.

Which row gives the correct experimental evidence for these properties?

	spontaneous nature of decay	random nature of decay
A	the decay rate is not affected by pressure	the decay rate is not affected by temperature
B	the decay rate is not affected by temperature	the rate at which radiation is received at a counter fluctuates
C	the rate at which radiation is received at a counter fluctuates	the decay rate is not affected by temperature
D	the decay rate is not affected by temperature	the decay rate is not affected by pressure

31. **[RVHS/H2/Prelims 2018/P1/30]**

Uranium $^{238}_{92}U$ decays to form lead $^{206}_{82}Pb$ by a series of alpha and beta emissions.

For each decay of uranium $^{238}_{92}U$ to lead $^{206}_{82}Pb$, how many α–particles and β–particles are emitted?

	α-particles	β-particles
A	6	2
B	6	8
C	8	4
D	8	6

32. [SRJC/H2/Prelims 2018/P1/29]

Which of the following is a correct description of mass defect?

A The difference between the mass of the nucleus of the products and reactants in a nuclear reaction.

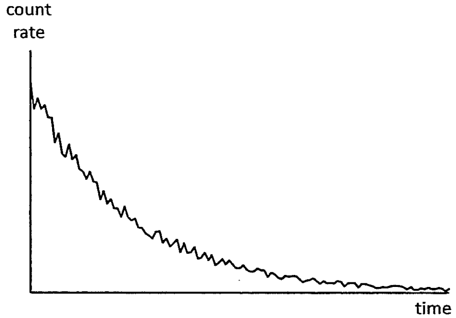
B It is the difference between the total mass of the neutrons and the mass of the nucleus.

C It is equal to the energy gained when individual nucleons comes together to form a nucleus.

D It is the binding energy of a nucleus divided by square of the speed of light.

33. [SRJC/H2/Prelims 2018/P1/30]

The graph below shows the variation of count rate from a particular radioactive sample with time.



What does the jagged feature of the graph indicate?

A It indicates the presence of background radiation.

B It indicates that the decay obeys radioactive decay law.

C It indicates the spontaneous nature of the radioactive decay.

D It indicates the random nature of the radioactive decay.

34. [TPJC/H2/Prelims 2018/P1/29]

Three different radioactive nuclides P, Q and R each decay by three different successive emissions.

P emits α α β

Q emits α β β

R emits β β β

Which nuclide produces a final nucleus that has the same proton number as its starting nucleus and which the same nucleon number as its starting nucleus?

	same proton number	same nucleon number
A	P	Q
B	P	R
C	Q	P
D	Q	R

35. [TPJC/H2/Prelims 2018/P1/30]

Nuclei of atoms can exist in excited states. When an excited nucleus returns to its state of lowest energy (the ground state), a γ-ray photon may be emitted.

The mass of a nucleus in its ground state is 59.9308 u. The energy of the photon emitted when this nucleus returns from an excited state to the ground state is 2.13×10^{-13} J.

What is the mass of the nucleus in the excited state?

A 59.9280 u B 59.9294 u C 59.9322 u D 59.9337 u

36. [VJC/H2/Prelims 2018/P1/29]

Which of the following statements concerning nuclear properties is true?

A The greater the binding energy of a nucleus, the more stable it is.

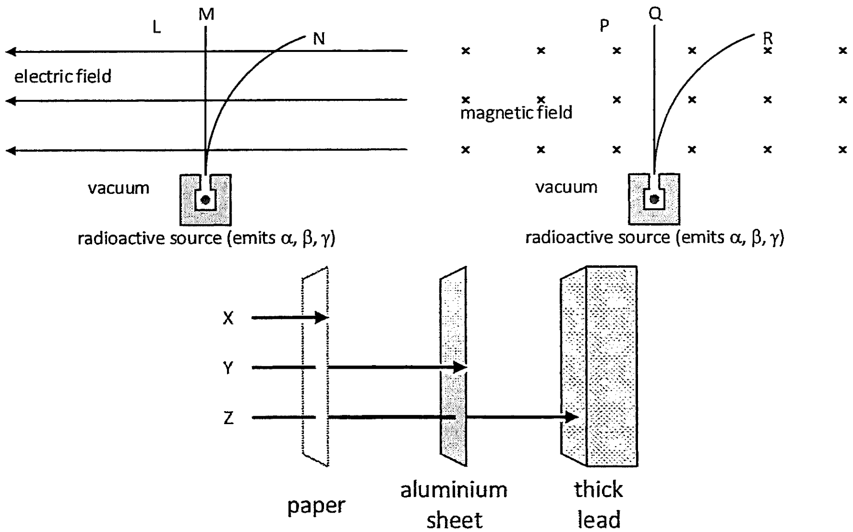
B If the total rest mass of the products of a reaction is greater than the total rest mass of the reactants, this reaction is impossible.

C The half-life of a radioactive substance can be changed by allowing the substance to react chemically to produce a new radioactive compound.

D When a stationary nucleus decays by emitting a γ-photon, the nucleus will move off in an opposite direction to the photon.

37. [VJC/H2/Prelims 2018/P1/30]

A radioactive source emits alpha, beta and gamma radiation. The three diagrams below illustrate the behaviours of the three radiations from this source.

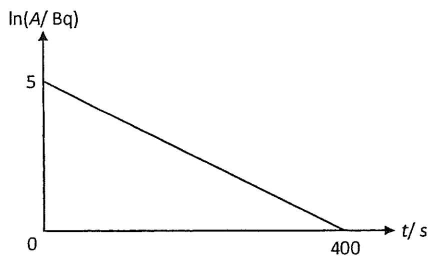


Which three labels refer to the same kind of radiation?

A L, R, X B M, Q, X C L, P, Z D N, R, Y

38. **[YJC/H2/Prelims 2018/P1/30]**

The graph below shows how the natural logarithm of the activity *A* of a radioactive isotope varies with time *t*.



What is the half-life of the isotope?

A 200 s B 55 s C 24 s D 0.42 s

II. Structured Questions (ACJC –YJC)

1. **[ACJC/H2/Prelims 2018/P2/6]**

a. Rutherford's α-particle scattering experiment was conducted to probe the structure of the atom.

Explain why only a small proportion of the α–particles incident on the gold foil are deflected through angles greater than 90° . **[3]**

b. A deuterium nuclear fusion reaction is given below:

$$^{2}_{1}H + ^{2}_{1}H \rightarrow A + B$$

where A is a charged particle and B is an unknown element.

The process is followed by a reaction involving *B* with another deuterium atom given below:

$$^{2}_{1}H + B \rightarrow ^{4}_{2}He + ^{1}_{0}n$$

Data:

Mass of deuterium $^{2}_{1}H$ atom: 2.01410 *u*

Mass of protium $^{1}_{1}H$ atom: 1.00783 *u*

Mass of neutron $^{1}_{0}n$: 1.00867 *u*

i. Deduce the nucleon and proton numbers of A and B. **[2]**

ii. Explain why the above nuclear fusion reactions cannot occur with slow moving particles of deuterium nuclides. **[1]**

iii. Determine the binding energy per nucleon of a $^{2}_{1}H$ nuclide, in MeV. **[4]**

c. Sodium-24, a radioactive isotope with a half-life of 15 hours, is used for the study of electrolytes within the human body.

A small volume of a solution containing sodium-24 had an activity of 12000 disintegrations per minute when it was injected into the bloodstream of a patient. After 39 hours, the activity of 1.0 cm^3 of the blood was found to have 0.50 disintegrations per minute.

i. Explain what is meant by the *half-life* of sodium-24. **[1]**

ii. Determine the volume of blood in the patient. **[3]**

iii. Suggest why sodium-24 is suitable for use in human patients. **[1]**

[Total: 15]

2. **[AJC/H2/Prelims 2018/P2/6]**

a. A sample of water is contaminated with radioactive iodine-131 ($^{131}_{53}I$).

The activity of the iodine-131 in 1.0 kg of this water is 460 Bq.

The half-life of iodine-131 is 8.1 days.

i. Calculate the number of iodine-131 atoms in 1.0 kg of this water. **[2]**

ii. An amount of 1.0 mol of non-contaminated water has a mass of 18 g.

Calculate the ratio

$$\frac{\text{number of molecules of water in 1.0 kg of contaminated water}}{\text{number of atoms of iodine-131 in 1.0 kg of contaminated water}}$$ **[2]**

iii. An acceptable limit for the activity of iodine-131 in water has been set as 170 Bq kg^{-1}.

Calculate the time, in days, for the activity of the contaminated water to be reduced to this acceptable level. **[2]**

b. A milk sample is to be tested for evidence of radioactive contamination with the radioactive nuclide strontium-90 (^{90}Sr), using a Geiger-Muller tube. The two stages involved in the decay of ^{90}Sr are described by the following two equations.

Stage 1: $^{90}_{38}\text{Sr} \rightarrow ^{90}_{39}\text{Y} + \beta$

Stage 2: $^{90}_{39}\text{Y} \rightarrow ^{90}_{40}\text{Zr} + \beta$

At each stage, a β-particle is emitted. At the end of stage 2, the product zirconium-90 (^{90}Zr) is stable

i. For stage 1, the emitted β-particles have a range of energies up to a maximum of 0.55 MeV.
Use conservation laws to explain why this range of energies leads to the suggestion that another particle is emitted by the decaying strontium-90 nucleus together with the β-particle. **[4]**

ii. Explain why it is difficult to measure the half-life of the beta decay of the strontium 90 experimentally. **[1]**

[Total: 11]

3. [CJC/H2/Prelims 2018/P3/6]

a. In 1919, Rutherford performed the first nuclear reaction induced in a laboratory in which a stationary nitrogen nucleus bombarded with an α-particle of a certain energy, transmutes to an oxygen nucleus and a proton.

Data:

Mass of nitrogen-14 nucleus: 13.9993 *u*

Mass of oxygen-17 nucleus: 16.9947 *u*

Mass of a proton : 1.0073 *u*

Mass of an α-particle: 4.0015 *u*

i. Write an equation for this nuclear reaction, showing the mass numbers and the atomic numbers of the particles involved. **[2]**

ii. Calculate the minimum kinetic energy of the alpha particle for the reaction to make this reaction occur. **[3]**

b. A radioactive isotope of thallium $^{207}_{81}$Tl emits a β-particle and is thought to emit a gamma photon. The half-life of $^{207}_{81}$Tl is 135 days.

i. The radiation is allowed to pass though perpendicularly a vertical uniform magnetic field and the photographs of traces is obtained in a cloud chamber under certain conditions. Fig 3.1 show tracks produced by the beta-particles and gamma ray photons.

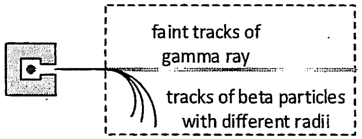


Fig. 3.1

Explain the features of the tracks. **[3]**

ii. An isotope of thallium emits a β-particle with an average energy of 2.4×10^{-13} J. Calculate

1. the total energy available from 1 g of thallium-207, **[2]**
2. the initial rate at which the β-particles are emitted from 1 g of the freshly prepared isotope, **[2]**
3. the initial power available from the beta particles emitted at the rate calculated in **b.ii.2.** **[1]**

[Total: 13]

4. [DHS/H2/Prelims 2018/P3/8]

a. Fig. 4.1 shows an α-particle A as it approaches and passes by a stationary gold nucleus N.

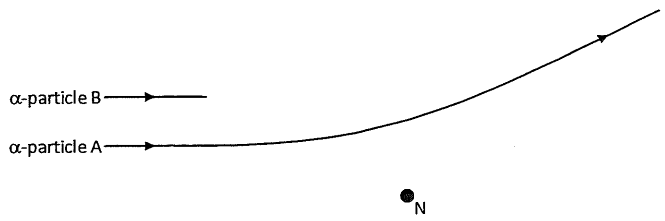


Fig. 4.1

A second α-particle B has the same initial direction and energy as α-particle A.

On Fig. 4.1, complete the path of α-particle B as it approaches and passes by the nucleus N. **[2]**

b. An alpha particle has a speed of 1.30×10^{7} m s^{-1}.

i. Calculate the kinetic energy of the alpha particle. **[2]**

ii. The alpha particle is aimed directly at a gold nucleus which has a proton number of 79.
Calculate the distance of closest approach *r*. **[3]**

c. A radiation detector is placed close to a radioactive source. The detector does not surround the source.
Radiation is emitted in all directions and, as a result, the activity of the source and the measured count rate are different.
Suggest two other reasons why the activity and the measured count rate may be different. **[2]**

d. The variation with time *t* of the measured count rate in **c.** is shown in Fig. 4.2.

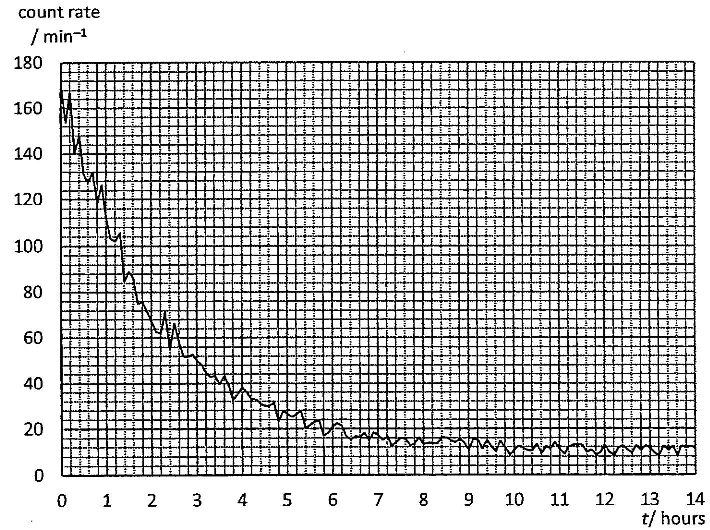


Fig. 4.2

i. State the feature of Fig. 8.2 that indicates the random nature of radioactive decay. [1]

ii. Use Fig. 4.2 to determine the half-life of the radioactive isotope in the source. [4]

e. The readings in **b.** were obtained at room temperature.

A second sample of this isotope is heated to a temperature of 500 °C.

The initial count rate at time t = 0 is the same as that in **b.**

The variation with time t of the measured count rate from the heated source is determined.

State, with a reason, the difference, if any, in

i. the half-life,

ii. the measured count rate for any specific time. [3]

f. A small volume of solution containing the radioactive isotope sodium-24 ($^{24}_{11}Na$) has an initial activity of 3.8×10^4 Bq. Sodium-24, of half-life 15 hours, decays to form a stable daughter isotope.

All of the solution is poured into a container of water. After 36 hours, a sample of water of volume $5.0\,cm^3$, taken from the container, is found to have an activity of 1.2 Bq.

Assuming that the solution of the radioactive isotope is distributed uniformly throughout the container of water, calculate the volume of water in the container. [3]

[Total: 20]

5. **[EJC/H2/Prelims 2018/P3/8 Part Modified]**

a. In an α-particle scattering experiment, an α-particle is travelling in a vacuum towards the centre of a gold nucleus, as illustrated in Fig. 5.1.

Fig. 5.1

The gold nucleus has a charge $79e$. At a large distance from the gold nucleus, the α-particle has energy 7.7×10^{-13} J.

i. The α-particle does not collide with the gold nucleus.

Show that the radius of the gold nucleus must be less than 4.7×10^{-14} m. [2]

ii. The results of the α-particle scattering experiment provide evidence for the structure of the atom.

Result 1: The vast majority of α-particles pass straight through the metal foil or are deviated by small angles.

Result 2: A very small minority of α-particles are scattered through angles greater than 90° and up to 180°.

State what may be inferred from

1. result 1, [1]
2. result 2. [2]

b. One nuclear reaction that can take place in a nuclear reactor may be represented, in part, by the equation

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{95}_{42}Mo + ^{139}_{57}La + 2^{1}_{0}n + \cdots + \text{energy}$$

Data for a nucleus and some particles are given in Fig. 5.2.

nucleus or particle	mass / u
$^{139}_{57}La$	138.955
$^{1}_{0}n$	1.00866
$^{1}_{1}p$	1.00728
$^{0}_{-1}e$	0.000549

Fig. 5.2

i. Complete the nuclear reaction shown above. [1]

ii. Calculate the binding energy per nucleon, in MeV, of lanthanum-139 $^{139}_{57}La$. [3]

iii. State and explain whether the binding energy per nucleon of uranium-235 ($^{235}_{92}U$) will be greater, equal to or less than your answer to **b.ii.** [2]

[Total: 11]

6. **[HCI/H2/Prelims 2018/P2/6]**

a. Fig. 6.1 shows the variation with the number of nucleons in the nucleus of binding energy per nucleon.

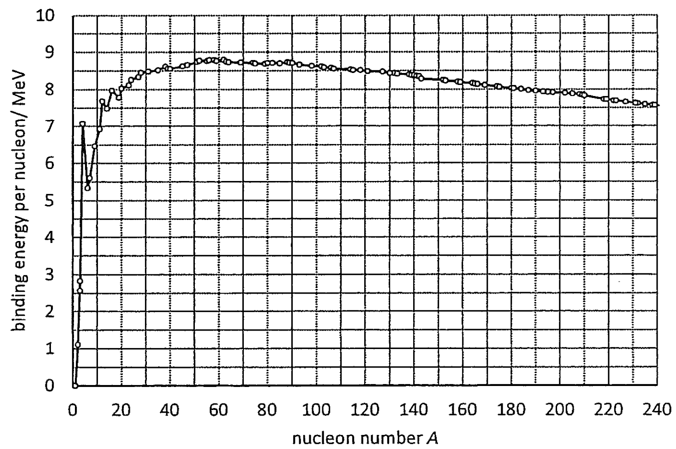


Fig. 6.1

i. Define binding energy of a nucleus. [1]

ii. Using Fig. 6.1, estimate the binding energy of the nucleus iridium-170, $^{170}_{77}Ir$. [1]

iii. Hence calculate the mass defect of $^{170}_{77}Ir$. [2]

b. Stellar nucleosynthesis is a collective term for nuclear reactions taking place in stars. These reactions create nuclei of elements heavier than hydrogen.

The "triple" alpha process is a nuclear fusion reaction that occurs in stars where three alpha particles combine to form carbon-12, ${}^{12}_{6}C$.

i. Determine the energy released in this reaction.

Data:

Mass of α-particle: 4.002603 *u*

Mass of ${}^{12}_{6}C$: 12.000000 *u* **[2]**

ii. "Silicon" burning is the final stage of fusion in massive stars. This process involves silicon-28, ${}^{28}_{14}Si$ capturing multiple alpha particles, until the sequence terminates at ${}^{56}_{28}Ni$. At this point the star can no longer release energy via nuclear fusion. This eventually results in a catastrophic collapse of the star.

Suggest a reason why the star can no longer release energy via nuclear fusion. **[1]**

[Total: 7]

7. [JJC/H2/Prelims 2018/P2/7]

a. In a nuclear reaction, uranium-235 (${}^{235}_{92}U$) nuclide is transformed into unstable uranium-236 nuclide (${}^{236}_{92}U$) through bombardment by a slow-moving neutron. The unstable uranium-236 nuclide undergoes nuclear fission to form stable products of lathium-139 nuclide (${}^{139}_{57}La$) and bromine nuclide (${}^{A}_{Z}Br$).

$${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{236}_{92}U \rightarrow {}^{139}_{57}La + {}^{A}_{Z}Br + 3{}^{1}_{0}n + \gamma$$

Data:

Rest mass of ${}^{235}_{92}U$ nuclide = 235.044u

Rest mass of ${}^{139}_{57}La$ nuclide = 138.906u

Rest mass of proton = 1.00728u

Rest mass of neutron = 1.00866u

i. Determine the nucleon number *A* and proton number *Z* of the bromine nuclide. **[1]**

ii. Calculate the binding energy of the uranium-235 nuclide, in MeV, to 4 significant figures. **[3]**

iii. State the feature of this equation that indicates that a chain reaction may be possible. **[1]**

b. In a laboratory source of strontium-90, the number of atoms present in the year 2013 was 2.36×10^{13}. Strontium-90 decays by emission of a β-particle and this nuclide has a half-life of 28 years.

i. State what is a β-particle. **[1]**

ii. Calculate the activity of the source in the year 2113. **[2]**

[Total: 8]

8. [MI/H2/Prelims 2018/P3/6]

The radioactive isotope plutonium-238 (${}^{238}_{94}Pu$) has a half-life of 86 years and decays by emitting an alpha particle with little or no gamma emission. The daughter nucleus is an isotope of uranium (U).

The mass of ${}^{238}_{94}Pu$ is 238.0496 *u* and the mass of an α-particle is 4.0026 *u*.

a. Write down an equation representing the decay, indicating clearly the atomic number and mass number of each nucleus. **[1]**

b. The total kinetic energy of the products is 5.649 MeV.

i. Calculate the mass of the uranium nucleus formed in the reaction, giving your answer in terms of atomic mass units to 4 decimal places. **[3]**

ii. By using the principle of conservation of momentum, show that the ratio of the kinetic energy of the alpha particle to that of the uranium nucleus is 58.5. **[3]**

iii. Hence, calculate the kinetic energy of the alpha particle. **[2]**

c. Satellites can be powered by the energy produced by the decay of plutonium. Plutonium is not placed in its pure form in the satellites, but installed as bricks of plutonium dioxide, PuO_2. PuO_2 is a ceramic and when it is shattered, it breaks into large pieces rather than smaller and more dangerous dust.

i. Explain why there is no difference in the probability of decay of plutonium whether it exists as pure plutonium or in the form of PuO_2. **[1]**

ii. Suggest a reason why large pieces of PuO_2 pose less of a health risk than plutonium dust, which can be inhaled into the body. **[1]**

[Total: 11]

9. [MJC/H2/Prelims 2018/P3/9 Modified]

A radon-222 (${}^{222}_{86}Rn$) nucleus, originally at rest, spontaneously decays to form a polonium-218 (${}^{218}_{84}Po$) nucleus and an alpha particle. It may be assumed that no gamma ray is emitted.

a. Explain what is meant by *spontaneous*. **[1]**

The rest masses of the nuclei are shown in Fig. 9.1.

${}^{222}_{86}Rn$	222.0176 u
${}^{218}_{84}Po$	218.0090 u
alpha particle	4.0026 u
proton	1.00728 u
neutron	1.00866 u

Fig. 9.1

b. i. Calculate the total kinetic energy of the decay products. **[2]**

ii. Describe the subsequent motion of the decay products. Explain your answer with reference to the principle of conservation of momentum. **[2]**

iii. Show that the ratio $\dfrac{\text{kinetic energy of alpha particle}}{\text{kinetic energy of Po-218 nucleus}} \approx 54.5$. **[2]**

c. i. Calculate the value of mass defect per nucleon for radon-222. Leave your answer in terms of atomic mass units *u*. **[3]**

ii. The mass defect per nucleon for polonium-218 has a value of $8.08312\times10^{-3}\,u$.

With reference to your answer in **c.i.**, explain whether Polonium-218 or Radon-222 is more stable. **[3]**

d. Radon-222 has a half-life of 3.8 days.

i. State what is meant by *half-life*. **[1]**

ii. Calculate the probability of a given radon-222 nucleus decaying per second. **[2]**

iii. A student stated that "radioactive materials with a short half-life always have a high activity". Discuss whether the student's statement is valid. **[1]**

e. A sample of radon-222 was carefully measured out and sealed in a container. The rate of radioactive decay was measured using an accurate instrument, taking into account background radiation. The number of alpha particles detected was significantly higher than expected.

State what this suggests about the stability of polonium-218. Explain your answer. **[3]**

[Total: 20]

10. [NJC/H2/Prelims 2018/P2/5]

a. This question is about an americium-241 radionuclide source used in a smoke detector.

A smoke detector works by detecting a decrease in the arrival of alpha particles from the americium when they are absorbed by the smoke. It is advised that the smoke detector be replaced after five years.

Data about the source:

Activity of source $=3.3\times10^4$ Bq

Decay constant $\lambda=4.8\times10^{-11}$ s^{-1}

i. Radioactive decay is random in nature. State what is meant by the word random in this context. **[1]**

ii. Use the equation $\Delta N=-\lambda N\Delta t$ with N taken as the original number of nuclei to show that about 5×10^{12} nuclei decay in the five years of use. **[2]**

iii. Explain why it is reasonable to use the equation $\Delta N=-\lambda N\Delta t$ (as above) to estimate the number of nuclei decaying over a five year period, but not over a few hundred years. **[2]**

iv. A decrease in activity of the sample is not likely to be the reason that the smoke detector should be replaced after five years. Suggest a more likely reason that the detector might need replacing. **[2]**

b. Strontium-90 is a radioactive waste product of nuclear power stations. Strontium-90 emits beta particles of energy up to 0.54 MeV. Should any of this isotope get into our food, it would be quickly absorbed into bones and teeth.

Explain why these beta particles are more of a risk to humans than gamma photons of the same energy. **[2]**

[Total: 9]

11. [NJC/H2/Prelims 2018/P3/8 Modified]

a. Describe how the alpha scattering experiment provides evidence for

i. the small size of the nucleus, **[1]**

ii. a charged nucleus. **[1]**

b. In 1914, James Chadwick showed that the energies of the β-particles emitted for a radioactive source had a distribution of energies rather than with a distinct single value of energy.

Figure 11.1 shows the energy spectrum for β-particles emitted during the decay of Bismuth-210 ($^{210}_{83}$Bi).

The intensity represents the number of β-particles emitted with a particular kinetic energy.

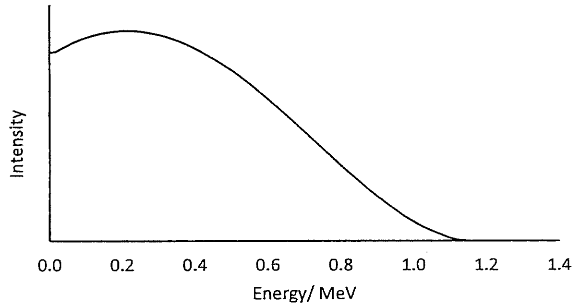


Fig. 11.1

i. 1. Determine, from Fig. 11.1, Q, the maximum possible energy of the beta particle emitted. **[1]**

2. Hence calculate the maximum speed of the beta particle. **[1]**

3. Comment on the value you obtained in **b.i.2.** **[1]**

ii. The radioactive isotope of Bismuth, $^{210}_{83}$Bi, decays into polonium (chemical symbol: Po) with the emission of a β-particle.

Determine the mass of the resultant Polonium nucleus, in terms of u, and express your answer to 3 decimal places. **[3]**

Data: Mass of $^{210}_{83}$Bi nucleus: 209.939 u

Mass of proton: 1.00729 u

Mass of neutron: 1.00867 u

iii. From Fig. 8.1, identify the most probable energy for the β-particle. **[1]**

iv. It is noted that the stable isotopes of heavy elements have an optimal neutron to proton ratio. Unstable isotopes will undergo transmutation into another element through radioactive decay such that product achieve the optimal ratio.

Suggest, with a reason, whether bismuth-210 has an excess of neutrons or protons, as compared to the optimal ratio. **[2]**

v. The continuous spectrum of kinetic energy values of the β-particle presented a problem to physicists up to 1930s. If a stationary nucleus decayed into a beta particle and a stable daughter nucleus only, it should lead to a distinct single value of energy.

Explain, using conservation of linear momentum and energy, how the continuous spectrum of β-particle energies gave rise to this problem. **[2]**

vi. Suggest what was proposed by physicists to resolve the problem in **v.** **[1]**

c. In an experiment, a detector is held a fixed distance from a sample of a radioactive material and the data provided is used to plot a graph of count-rate against time, as shown in Fig 11.2.

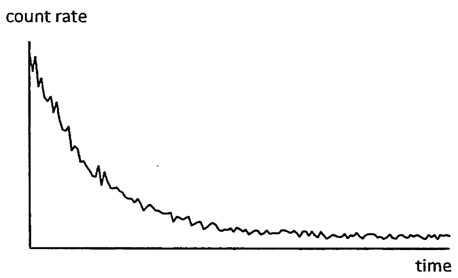


Fig. 11.2

i. Explain why the data points on the graph do not lie on a smooth curve. [1]

ii. Suggest two reasons why the count-rate recorded is not equal to the activity of the radioactive sample. [2]

iii. The count rate from a counter near a radioactive source is $7.6\times10^{8}\ s^{-1}$. The decay constant of the source is $4.6\times10^{-3}\ s^{-1}$.

1. Calculate the time taken for the count rate to fall to $8.3\times10^{3}\ s^{-1}$. [1]
2. When the detector is at a distance y from the radioactive source, the count rate is 234 counts per minute. Calculate the count rate when the detector is at a distance $3y$ from the source. [2]

[Total: 20]

12. [NYJC/H2/Prelims 2018/P2/5]

A stationary copper-64 nucleus undergoes a nuclear reaction to form a zinc-64 nucleus and a beta-particle. After the reaction, the zinc-64 nucleus moves off with negligible speed, while the beta-particle moves off with high speed.

a. Explain qualitatively why the beta-particle moves off with a much higher speed than the zinc-64 nucleus. [2]

b. The kinetic energy E_K and the distance r of the β-particle from the centre of zinc nucleus are monitored. Fig. 12.1 shows the variation with $1/r$ of the values of E_K.

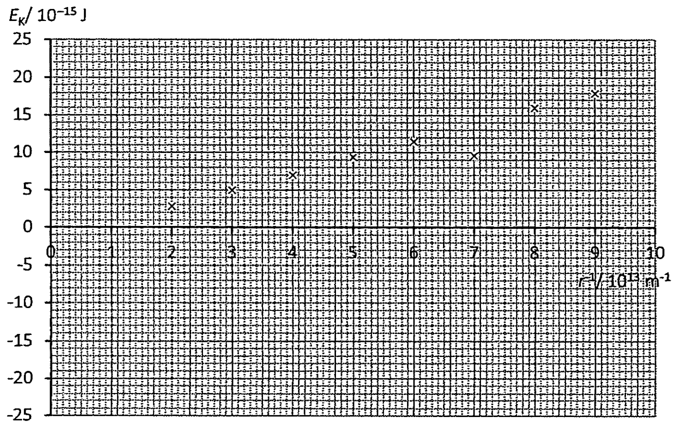


Fig. 12.1

Assume that the beta-particle and zinc nucleus form an isolated system.

i. By drawing an appropriate line on Fig. 12.1, determine the y-intercept, E_0, of the graph. [2]

ii. Explain the physical significance of the value of E_0 found in **b.i.** [2]

iii. Assuming that total energy of the beta-particle remains constant during its motion, sketch labelled graphs on the axes in Fig. 12.1 to represent

1. the variation of the total energy E_T.
2. the variation of the electric potential energy E_P.

[Total: 9]

13. [NYJC/H2/Prelims 2018/P3/9]

a. Fig. 13.1 below shows a Rutherford scattering experiment in which α-particles are directed at a gold foil. The detector is shown in two positions in the evacuated chamber.

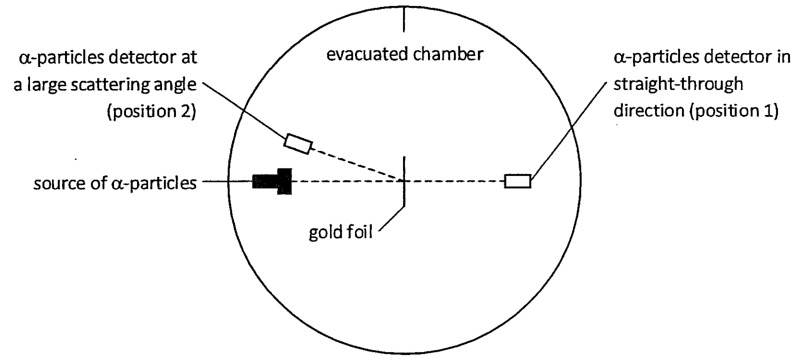


Fig. 13.1

i. Explain why air needs to be removed from the apparatus. [1]

ii. Explain why the gold foil should be very thin. [1]

iii. State what can be deduced, from the following observations, about the structure of the atom and the properties of the gold nucleus.

1. A high count rate is detected by the α-particle detector in position 1. [1]
2. A low count rate is detected by the α-particle detector in position 2. [2]

b. Fig. 13.2 illustrates the variation with nucleon number of the binding energy per nucleon of nuclei.

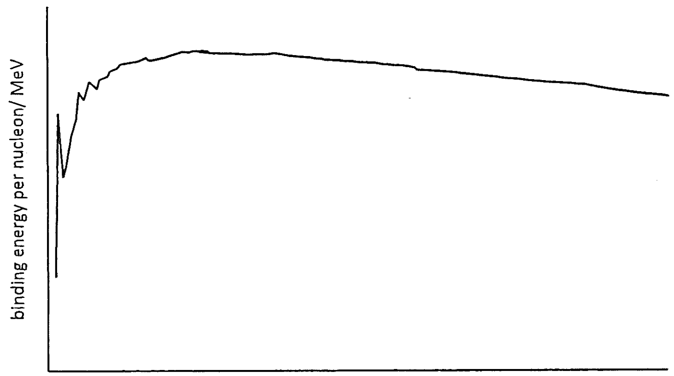


Fig. 13.2

i. Explain what is meant by the *binding energy* of a nucleus. [2]

ii. With the aid of Fig. 9.2, explain why more energy per nucleon is released in fusion than in fission. [3]

iii. On Fig. 13.2, mark with the letter **S** the region of the graph representing nuclei having the greatest stability. [1]

iv. On Fig. 13.2, complete the graph for elements of low nucleon numbers. [1]

c. Uranium-235 may undergo fission when bombarded by a slow-moving neutron to produce xenon-142 and strontium-90 as shown below.

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{142}_{54}Xe + ^{90}_{39}Sr + \text{neutrons} + ^{0}_{-1}e$$

i. Determine the number of neutrons produced in this fission reaction. [1]

ii. State the role of these neutrons in the fission reaction. [1]

iii. Data on binding energy per nucleon for some nuclides is given in Fig. 13.3.

Isotope	Binding energy per nucleon / MeV
Uranium-235	7.59
Xenon-142	8.23
Strontium-90	8.70

Fig. 13.3

1. the energy released in this fission reaction, [2]
2. the mass equivalent of this energy. [1]

iv. A nuclear power station supplies 100 MW of electrical power to a town for one year at an efficiency of 35%. Calculate the mass of uranium needed to operate the power station for one year. [3]

[Total: 20]

14. [PJC/H2/Prelims 2018/P3/10 Modified]

a. Define half-life. [1]

b. The isotope ^{59}Fe is a β-emitter with a half-life of 44.6 days. In a medical investigation of blood disorders, a dose of ^{59}Fe with an initial activity of 8.50×10^{7} Bq is injected into the blood stream of a patient.

i. Calculate

1. the probability per second of the decay of a ^{59}Fe nucleus, [2]
2. the initial number of ^{59}Fe nuclei injected into the body, [2]
3. the time taken for the activity to be one-fifth of the initial value. [2]

ii. Explain why in practice, the time taken for the activity in the patient to become one-fifth of the initial value is shorter than that calculated in **i.3.** [1]

c. i. On Fig. 14.1, sketch a graph to show the variation with nucleon number *A* of the binding energy per nucleon. Label in the graph an approximate value, in MeV, of the maximum binding energy per nucleon. [2]

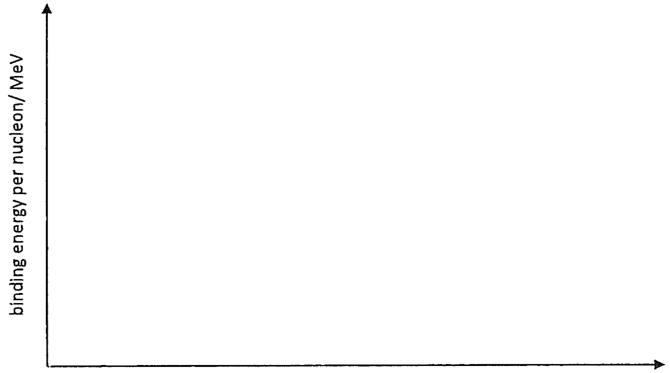


Fig. 14.1

ii. Using Fig. 14.1, explain why energy is released in a fusion reaction. [2]

iii. A possible nuclear reaction which has been suggested for the generation of energy is represented by the following equation:

$$^{2}_{1}H + ^{3}_{1}H \rightarrow ^{4}_{2}He + X$$

Data for the above reaction are given in Fig. 14.2.

Particle	Mass/ *u*
X	1.008665
$^{2}_{1}H$	2.014102
$^{3}_{1}H$	3.016050
$^{4}_{2}He$	4.002603

Fig. 14.2

1. Identify the nucleus represented by the symbol *X*. **[1]**
2. Using the data in Fig. 14.2, determine the energy produced by 150 kg of an appropriate mixture of the hydrogen isotopes. **[4]**
3. The average power needed by a nuclear submarine is 110 MW and the efficiency of the above process is 15%.

   Determine the length of time, in years, that 150 kg of the mixture can supply sufficient energy to the submarine. **[3]**

[Total: 20]

15. [RVHS/H2/Prelims 2018/P3/6 Modified]

The table shows the half-life of various radioactive nuclides.

nuclide	$^{197}_{84}\text{Po}$	$^{220}_{86}\text{Rn}$	$^{222}_{88}\text{Ra}$	$^{234}_{91}\text{Pa}$
half-life/ s	60	52	38	70

The variation with time of the number of nuclei for one of these radioactive nuclides is shown in Fig. 15.1.

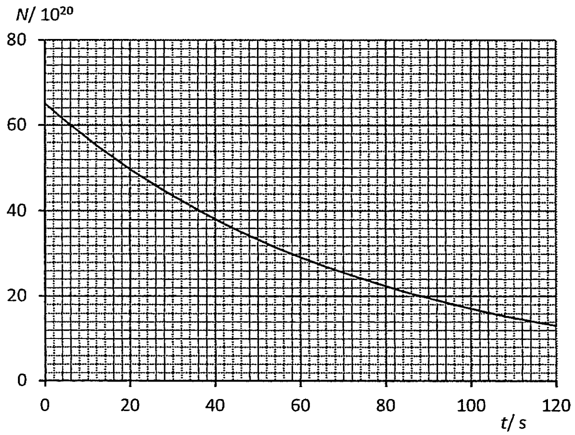


Fig. 15.1

a. i. Using Fig. 15.1, determine which is the decaying radioactive nuclide. Show your working clearly. **[3]**

ii. Hence, calculate the initial activity of the radioactive nuclide. **[2]**

The decaying radioactive nuclide emits an alpha particle as it decays into a stable daughter nuclide Y.

b. By plotting 3 points on Fig. 15.1, sketch a graph of the variation with time of number of daughter nuclide Y. Show your working clearly. **[2]**

c. Write a complete nuclear equation for the decay process. **[1]**

d. Given that linear momentum is conserved in this decay, show that the ratio

$$\frac{\text{initial kinetic energy of } \alpha\text{-particle}}{\text{initial kinetic energy of daughter nucleus Y}} = \tfrac{1}{4}A - 1$$

where *A* is the nucleon number of the decaying radioactive nuclide. **[2]**

[Total: 10]

16. [SRJC/H2/Prelims 2018/P2/6 Modified]

a. A stationary nucleus of a radioactive nuclide, $^{218}_{84}\text{Po}$, underwent a chain of decays by the emission of an α and β-particles. The decay is represented by the two equations:

$$^{218}_{84}\text{Po} \rightarrow \text{D} + \alpha$$

$$\text{D} \rightarrow \text{E} + \beta$$

where D and E are the nuclides formed after the decay.

i. State the nuclear notation for E. **[1]**

ii. Determine the ratio of the kinetic energy of the α-particle to the total kinetic energy of D and α-particle. **[3]**

iii. In reality, β-particles have a range of kinetic energies, instead of a fixed value. Explain how this is possible. **[1]**

iv. A sample of $^{218}_{84}\text{Po}$ is placed on a weighing balance and a reading of 4.05 g is obtained. After 243 s, the reading drops to 4.02 g.

1. Determine the number of particles $^{218}_{84}\text{Po}$ in the initial sample. **[1]**
2. Show that the total number of particles D and E after 243 s is 4.52×10^{21}. **[1]**
3. Determine the half-life of $^{218}_{84}\text{Po}$. **[2]**

b. For many unstable parent nuclei, the daughter nuclei is itself radioactive. This may give rise to a radioactive series where there may be ten or more different radioactive daughter products.

A radioactive parent nucleus X has a radioactive daughter nucleus Y and, in turn, this daughter produces a further stable daughter Z.

The variation with time *t* of the percentage number *P* of the different nuclei X, Y and Z in a radioactive sample is illustrated in Fig. 16.1.

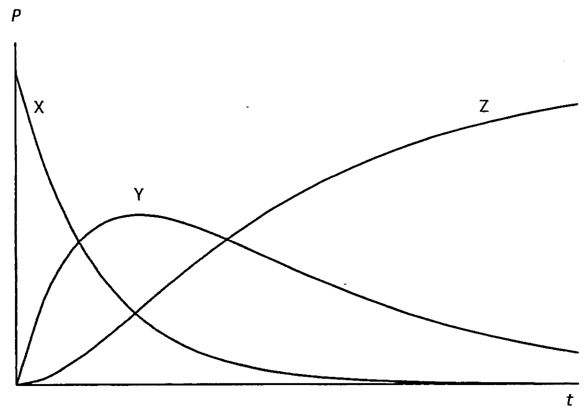


Fig. 16.1

i. X has a half-life of 5 hours. The count rate at $t = 0$ and $t = 5$ hours were measured.

Suggest three possible reasons why the count rate is not exactly halved after 5 hours. **[3]**

ii. Explain why graph of Y increases to a maximum and then decreases. **[2]**

[Total: 14]

17. [SRJC /H2/Prelims 2018/P3/7]

An induced nuclear fission reaction may be represented by the equation

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + 3^{1}_{0}n$$

a. Sketch the variation with nucleon number of the binding energy per nucleon in Fig 17.1. [2]

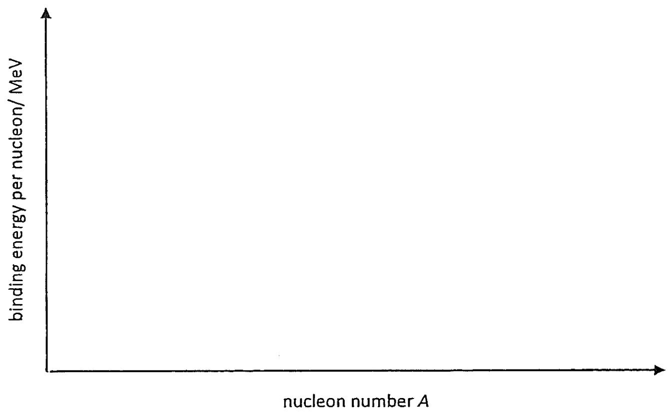


Fig. 17.1

b. Hence, explain why Uranium – 235 is more likely to undergo fission than fusion. [3]

c. The masses of the various nuclides are as listed below.

nuclide	mass/ *u*
Uranium-235	235.04393
Barium-141	140.91441
Krypton-92	91.92616
Neutron	1.00866

Determine the energy released in the reaction. [2]

[Total: 7]

18. [TPJC/H2/Prelims 2018/P3/8 Modified]

a. Explain what is meant by

i. an isotope. [1]

ii. radioactive decay. [2]

iii. half-life of a radioactive substance. [2]

b. The number of atoms present in a radioactive source when it was first acquired in 1990 was 2.5×10^{14}.

In 2018, the number of atoms that remains is 1.3×10^{14}.

Determine the number of atoms that will remain in 2032. [4]

c. Strontium-90 ($^{90}_{38}Sr$) undergoes β-decay to form yttrium (Y).

Yttrium is itself radioactive and decays by the emission of a β-particle. The yttrium decays to form zirconium (Zr) which is stable.

Complete the nuclear equations below for the decay of strontium-90 to form zirconium. [2]

d. Plutonium-239 decays by α-emission to form the isotope uranium-235. In this nuclear reaction, 5.26 MeV of energy is released. The energy of the emitted α-particle is 5.15 MeV.

Suggest and explain why these two values are different. [2]

e. Fig. 18.1 lists various nuclides and their respective mass and binding energy per nucleon.

Particle/ Nuclide	Name	Mass/ *u*	Binding energy per nucleon/ MeV
$^{1}_{1}H$	Proton	1.00728	0
$^{1}_{0}n$	Neutron	1.00866	0
$^{4}_{2}He$	Helium	-	7.07470
$^{14}_{7}N$	Nitrogen	-	7.47724
$^{17}_{8}O$	Oxygen	-	7.75224

Fig. 18.1

i. Explain why the binding energy of a proton or neutron is zero. [1]

ii. Using the values from Fig. 18.1, determine the mass of helium nucleus, expressing your answer to an appropriate number of significant figures. [3]

iii. In a particular nuclear reaction, a slow moving alpha particle bombards a stationary nitrogen nucleus $^{14}_{7}N$, transmuting it to an oxygen nucleus $^{17}_{8}O$ and a proton.

The nuclear reaction is shown below.

$$^{14}_{7}N + ^{4}_{2}He \rightarrow ^{17}_{8}O + ^{1}_{1}H$$

Making use of values from Fig. 18.1, determine quantitatively whether the reaction can be spontaneous. Explain your workings. [3]

[Total: 20]

19. [VJC/H2/Prelims 2018/P2/6 Modified]

Radioactive decay is the *spontaneous* and random emission of radiation from a radioactive source.

Radioactive nuclide tellurium-131 ($^{131}_{52}Te$) decays by β emission to form iodine-131 ($^{131}_{53}I$).

Iodine-131 is not stable, and decays by β emission to the stable isotope xenon-131 ($^{131}_{54}Xe$).

The *half-life* for this second decay is very much longer than that for the decay of $^{131}_{52}Te$.

A sample of pure $^{131}_{52}Te$ is prepared at time $t = 0$. Fig 19.1 shows the activity A of the sample over a period of 10 hours.

time t/ hr	activity A/ 10^9 Bq
0	1110
1	212
2	42.1
3	9.91
4	2.64
5	2.41
6	2.36
7	2.35
8	2.35
9	2.35
10	2.35

Fig. 19.1

a. Explain the meaning of the terms spontaneous and half-life. **[2]**

b. The relation between activity A and time t follows the expression where A_0 is the initial activity and λ is a constant.

$$A = A_0 e^{-\lambda t}$$

Data from Fig. 19.1 is used to obtain values for $\ln A$.
The variation with time t of $\ln A$ is plotted on the graph of Fig 19.2.

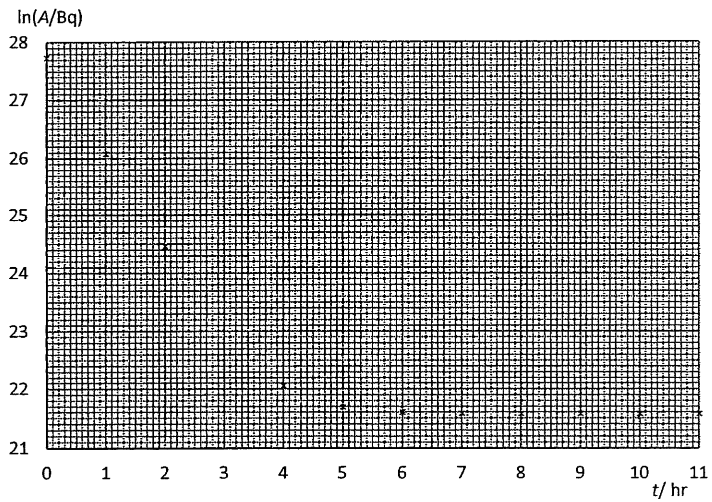


Fig. 19.2

i. Determine $\ln A$ for $t = 3$ hr. **[1]**

ii. On Fig. 19.2,

1. plot the point corresponding to $t = 3$ hr
2. draw the line of best fit for all the points. **[2]**

iii. Explain the shape of the line drawn in **ii.** **[2]**

iv. Using the line drawn in **ii.**, determine the decay constant of $^{131}_{52}Te$. **[2]**

v. Hence, or otherwise, deduce the initial number of $^{131}_{52}Te$ nuclei. **[2]**

vi. Explain why the amount of $^{131}_{53}I$ in the sample first increases and then decreases. **[2]**

vii. Assume that all the $^{131}_{52}Te$ has been converted to $^{131}_{53}I$ after about 6 hours. Calculate the half-life of $^{131}_{53}I$. **[3]**

[Total: 16]

20. **[VJC/H2/Prelims 2018/P3/6 Modified]**

Fig. 20.1 shows the variation with nucleon number A of binding energy per nucleon for nuclides with a nucleon number greater than 40.

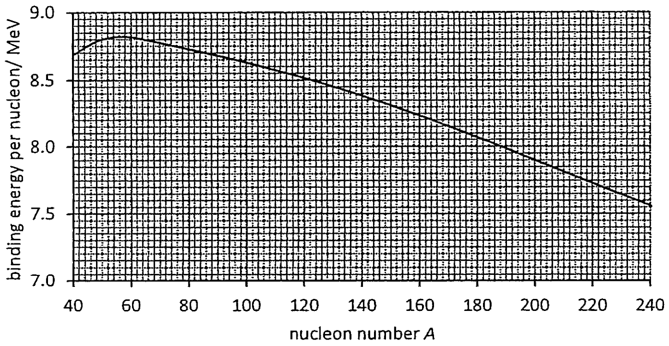


Fig. 20.1

The following nuclear reaction occurs when a slow moving neutron is absorbed by an isotope of uranium 235.

$$^{235}_{92}U + ^{1}_{0}n \rightarrow ^{141}_{56}Ba + ^{92}_{36}Kr + \text{neutrons}$$

a. Determine the number of neutrons produced in this reaction. **[1]**

b. Explain how this reaction is able to produce energy. **[2]**

c. Use the graph in Fig. 20.1 to determine the energy released in the above reaction. **[3]**

d. Suggest why the neutron is not included in the calculation above. **[1]**

[Total: 7]

21. **[YJC/H2/Prelims 2018/P3/9 Part]**

c. The decay of a calcium-45 nucleus (chemical symbol: Ca; $Z = 20$) releases a scandium-45 (chemical symbol: Sc; $Z = 21$) nucleus. Write a nuclear equation to deduce the type of decay that takes place. **[2]**

d. Radon-219 is a naturally radioactive element, a member of the noble gas family found in soil.

Data for the α-decay of Radon-219 to form Polonium-215 are given in Fig. 21.1 below.

Nucleus	Mass of nucleus / u
Radon-219	219.009523
Polonium-215	214.999469
Helium-4	4.002603

Fig. 21.1

Determine the energy released, in MeV, during the decay. **[3]**

e. Bismuth-210 undergoes β-decay. The emitted β-particles have a range of energies up to a maximum of 1.17 MeV.

Using conservation laws, explain why this range of energies leads to the suggestion that, apart from the β-particle, another particle must be emitted by the Bismuth-210 nucleus. **[4]**

[Total: 9]